PROBLEM SET 1, PART 1: SMOOTH MANIFOLDS
DUE: SEP. 26

(1) [**Topological group**]
A topological group is a topological space G which is also a group, so that the multiplication operation

$$\mu : G \times G \to G, \quad (g_1, g_2) \mapsto g_1 g_2$$

and the inverse operation

$$i : G \to G, \quad g \mapsto g^{-1}$$

are both continuous. For any subsets S and T in G we can define

$$ST = \{g_1 g_2 \mid g_1 \in S, g_2 \in T\}.$$

By this way we can define the subsets $S^n = S \times S \times \cdots \times S$ and S^{-1}.

(a) If G is a topological group, and U is any open neighborhood of the identity element $e \in G$. Prove: There exists an open neighborhood V of e so that $V = V^{-1}$ and $V^2 \subset U$.

(b) Prove: If G is a connected topological group, then for any open neighborhood U of the identity element $e \in G$, we have

$$G = \bigcup_{n=1}^{\infty} U^n.$$

(c) [Not required] Suppose G is compact Hausdorff topological group and $g \in G$. Prove: $e \in \overline{\{g^n \mid n \in \mathbb{Z} \setminus \{0\}\}}$.

(2) [**Locally Euclidean**]
Prove the following properties of locally Euclidean spaces:

(a) Any connected component of a locally Euclidean space is open.

(b) Any connected locally Euclidean space is path connected.

(c) [Not required] Any locally Euclidean Hausdorff space is regular. [Thus as a consequence of Urysohn's metrization theorem, any topological manifold is metrizable.]

(d) [Not required] If both X, Y are connected, second countable and locally Euclidean, and $f : X \to Y$ is bijective and continuous, then f is a homeomorphism.

(3) [**Topological manifolds with boundary**]

(a) Find the definition of topological manifolds with boundary from literature.

(b) Prove: If M is a topological n-manifold with boundary, then its boundary, ∂M, is a topological $(n-1)$-manifold without boundary.

(c) [Not required] Prove: the product of two topological manifolds with boundaries is a topological manifold with boundary. What is its boundary?

(4) [**Connected topological manifolds are homogeneous**]
Let M be a connected topological manifold. Prove: for any $p, q \in M$, there exists a homeomorphism $\varphi : M \to M$ so that $\varphi(p) = q$.

(5) [**Local homeomorphism**]
Let X, Y be topological spaces. A map $f : X \to Y$ is called a *local homeomorphism* if for every point $x \in X$, there exists an open set U containing x such that the image $f(U)$ is open in Y, and the restriction $f|_U : U \to f(U)$ is a homeomorphism (with respect to the respective subspace topologies).
   (a) Show that every local homeomorphism is an open map (i.e. maps each open set to an open set).
   (b) Show that if a local homeomorphism is bijective, then it is a homeomorphism.
   (c) Show that if Y is locally Euclidean and $f : X \to Y$ is a local homeomorphism, then X is locally Euclidean.
   (d) Show that if X is locally Euclidean and $f : X \to Y$ is a surjective local homeomorphism, then Y is locally Euclidean.

PROBLEM SET 1, PART 2: SMOOTH MANIFOLDS/MAPS/FUNCTIONS
DUE: SEP. 26

(1) [**Construct smooth manifolds by gluing Euclidean open sets**]
Let M be a smooth manifold with atlas $\mathcal{A} = \{(\varphi_\alpha, U_\alpha, V_\alpha)\}$.
(a) Prove: the transition maps $\varphi_{\alpha\beta} := \varphi_\beta \circ \varphi_\alpha^{-1}$ satisfy the *cocycle conditions*
(i) $\varphi_{\beta\gamma} \circ \varphi_{\alpha\beta} = \varphi_{\alpha\gamma}$ on $\varphi_\alpha(U_\alpha \cap U_\beta \cap U_\gamma)$.
(ii) $\varphi_{\alpha\alpha} = \mathrm{Id}_{V_\alpha}$.
(iii) $\varphi_{\alpha\beta} = (\varphi_{\beta\alpha})^{-1}$.
(b) Now on the disjoint union $\widetilde{M} := \bigsqcup_\alpha V_\alpha$ we define an equivalence relation via

$$x \sim y \iff \exists \alpha, \beta \text{ s.t. } x \in V_\alpha, y \in V_\beta \text{ and } y = \varphi_{\alpha\beta}(x).$$

(i) Check: $\sim$ is an equivalence relation on $\widetilde{M}$.
(ii) Prove: the quotient $\widetilde{M}/\sim$ is homeomorphic to M.
(iii) Define a natural smooth structure on $\widetilde{M}/\sim$

(2) [**Product of manifolds**]
(a) Prove: The product of two topological manifolds is a topological manifold.
(b) Let M, N be smooth manifolds with atlases $\{(\phi_\alpha, U_\alpha, V_\alpha)\}$ and $\{(\psi_\beta, X_\beta, Y_\beta)\}$ respectively. Define an atlas on the product space $M \times N$ to be $\{(\phi_\alpha \times \psi_\beta, U_\alpha \times X_\beta, V_\alpha \times Y_\beta)\}$. Check that this gives a smooth structure on $M \times N$.
(c) Explicitly construct local charts on $\mathbb{T}^2 = S^1 \times S^1 \subset \mathbb{R}^4$ to make it into a smooth manifold. (You need to prove that your charts are compatible.)

(3) [**The need of global continuity in the definition of smoothness**]
(a) Define two atlases $\mathcal{A}$ and $\mathcal{B}$ on $\mathbb{R}$ as follows:

$$\mathcal{A} = \{(\varphi, \mathbb{R}, \mathbb{R})\}, \quad \varphi(x) = x,$$
$$\mathcal{B} = \{(\psi_i, X_i, Y_i) : i \in \mathbb{Z}\}, \quad \text{where } X_i = Y_i = (i-1, i+1) \text{ and } \psi_i(x) = x.$$

Define a map $f : (\mathbb{R}, \mathcal{A}) \to (\mathbb{R}, \mathcal{B})$ via

$$f(x) = \begin{cases} 2, & x > 0 \\ 0, & x \le 0 \end{cases}$$

Prove: For each i, the map $\psi_i \circ f \circ \varphi^{-1} : \varphi(\mathbb{R} \cap f^{-1}(X_i)) \to \psi_i(X_i)$ is smooth. [So we can't drop the continuity assumption in Definition 1.1 in Lecture 4.]
(b) Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds. Suppose for each point p, there exists $(\varphi_\alpha, U_\alpha, V_\alpha) \in \mathcal{A}$ and $(\psi_\beta, X_\beta, Y_\beta) \in \mathcal{B}$ so that $p \in U_\alpha$ and $f(U_\alpha) \subset X_\beta$. Prove: Under this assumption we can drop the continuity assumption of f in Definition 1.1 in Lecture 4.

(4) [**Orientability of smooth manifolds**]
Let M be a smooth manifold of dimension n. We say M is *orientable* if there exists an atlas $\{(\varphi_\alpha, U_\alpha, V_\alpha)\}$ of M so that

$$\det(d\varphi_{\alpha\beta})_p > 0$$

for all $p \in \varphi_\alpha(U_\alpha \cap U_\beta)$, where $d\varphi_{\alpha\beta}$ is the Jacobian matrix of $\varphi_{\alpha\beta}$.

(a) Prove: S^n is orientable.
(b) Prove: For $\mathbb{RP}^n$, the atlas we constructed in Lecture 2 makes $\mathbb{RP}^n$ orientable for odd n but not for even n.
(c) [Not required] Prove: $\mathbb{CP}^n$ is orientable.
(d) Prove: If M, N are orientable, so is $M \times N$. Conversely, if $M \times N$ are orientable, so are M and N.

(5) [**Smooth Urysohn's Lemma**]
Suppose A and B are two disjoint closed subsets of a smooth manifold M.

(a) Show that there exists a smooth function f on M so that $0 \leq f(x) \leq 1$ for all $x \in M$, $f|_A = 0$ and $f|_B = 1$.
(b) [Not required] For any closed subset K in M, there is a smooth nonnegative function $f : M \to \mathbb{R}$ so that $f^{-1}(0) = K$.
[Hint: First work out this problem for $M = \mathbb{R}^n$, in which case $\mathbb{R}^n \setminus K$ can be written as the union of countably many open balls. Try to construct a smooth function on each such ball which is positive inside the ball and equals 0 outside the ball. Then add these functions (each multiplied by suitable factors to guarantee smoothness).]
(c) Show that there exists a smooth function f on M so that $0 \leq f(x) \leq 1$ for all $x \in M$, $f^{-1}(0) = A$ and $f^{-1}(1) = B$.

(6) [**Exhaustion function**]
A real-valued continuous function f on M is called an *exhaustion function* for M if for any $c \in \mathbb{R}$, the *sublevel set* $f^{-1}((-\infty, c])$ is compact. Prove: There exists a positive smooth exhaustion function on any smooth manifold.

PROBLEM SET 2, PART 1: THE DIFFERENTIAL
DUE: OCT. 17

(1) [Not required] [**Tangent vectors defined by curves**]
There is a more geometric way to define tangent vectors: Take a chart $\{\varphi, U, V\}$ around p. Let $\mathcal{C}_p$ be the set of all smooth maps $\gamma : (-\varepsilon, \varepsilon) \to U$ such that $\gamma(0) = p$. Define an equivalence relation on $\mathcal{C}_p$ by

$$\gamma_\alpha \sim \gamma_\beta \iff \frac{d(\varphi \circ \gamma_\alpha)}{dt}(0) = \frac{d(\varphi \circ \gamma_\beta)}{dt}(0).$$

Define a natural vector space structure on $\mathcal{C}_p/\sim$ so that T_pM is isomorphic to $\mathcal{C}_p/\sim$.

(2) [**Product of manifolds – continued**]
Let M, N be smooth manifolds and consider the product $M \times N$. Prove:
(a) The nature projections $\pi_1 : M \times N \to M$ and $\pi_2 : M \times N \to N$ are smooth.
(b) [Universality] If P is any smooth manifold, then a map $f : P \to M \times N$ is smooth if and only if both $\pi_1 \circ f$ and $\pi_2 \circ f$ are smooth.
(c) $T_{(p_1,p_2)}(M_1 \times M_2) \simeq T_{p_1}M_1 \oplus T_{p_2}M_2$.
(d) More generally, let $f_1 : M_1 \to N_1$ and $f_2 : M_2 \to N_2$ be smooth maps, and $p_1 \in M_1, p_2 \in M_2$. Find the relation between $d(f_1 \times f_2)_{(p_1,p_2)}$ and $(df_i)_{p_i}$, $i = 1, 2$.

(3) [**The tangent bundle**]
Let M be a smooth manifold of dimension n. Let $TM = \bigcup_p T_pM$ be the disjoint union of all tangent vectors. We will call TM the *tangent bundle* of M. We will denote an element X_p in TM by (p, X_p), to emphasis its "base point". There is a natural projection map

$$\pi : TM \to M, \quad (p, X_p) \mapsto p.$$

For each chart (φ, U, V) of M, we define a bijective map

$$T\varphi = (\varphi \circ \pi, d\varphi) : \pi^{-1}(U) \to V \times \mathbb{R}^n, \quad (p, X_p) \mapsto (\varphi(p), d\varphi_p(X_p)).$$

We endow a topology on TM so that each $T\varphi$ is a homeomorphism. Prove:
(a) $(T\varphi, \pi^{-1}(U), V \times \mathbb{R}^n)$ is a chart of TM.
(b) Charts of this type are compatible, so TM is a $2n$-dimensional smooth manifold.
(c) No mater M is orientable or not, the tangent bundle TM is orientable.
(d) The differential $d\pi_{(p,X_p)} : T_{(p,X_p)}TM \to T_pM$ is always surjective.
(e) TS^1 is diffeomorphic to $S^1 \times \mathbb{R}$.

(4) [Not required] [**Submersions are open maps**]
(a) Let $f : M \to N$ be a submersion. Prove: f is an open map.

(b) Prove: If M is compact and N is connected, then any submersion $f : M \to N$ is surjective. Conclude that there exists no submersion from any compact smooth manifold M to any connected noncompact smooth manifold (e.g. $\mathbb{R}^n$).

(5) [**Hadamard's global inverse function theorem**]
Let M, N be connected smooth manifolds, and $f \in C^\infty(M, N)$.
(a) Prove: If f is proper (i.e. the pre-images of compact subsets are compact), then it is closed (i.e. the image of closed subsets are closed).
(b) [Not required] Prove: If f is proper and local diffeomorphism everywhere, then it is a covering map.
(c) Finally conclude that if f is proper and local diffeomorphism everywhere, and N is simply connected, then f is a diffeomorphism.

(6) [**Compositions of submersion/immersions/constant rank maps**]
Consider the compositions of submersions/immersions/constant rank maps. [We assume all maps are smooth.]
(a) Is it true that the composition of two submersions is still submersion? What about the composition of immersions? What about constant rank maps?
(b) What if we composite a constant rank map with a submersion? With an immersion? A submersion with an immersion?
(c) If the composition $g \circ f$ is an immersion, can we conclude f is an immersion?
(d) If the composition $g \circ f$ is a submersion, can we conclude g is a submersion? If not, what extra condition do we need?

(7) [**Matrix Lie groups**]
Recall that $\mathrm{GL}(n, \mathbb{R})$ is an n^2 dimensional smooth manifold.
(a) Consider the map $\det : \mathrm{GL}(n, \mathbb{R}) \to \mathbb{R}$.
  (i) Prove: det is a smooth function.
  (ii) For any $X \in \mathrm{GL}(n, \mathbb{R})$, what is $T_X(\mathrm{GL}(n, \mathbb{R}))$?
  (iii) Show that $(d \det)_X(A) = (\det X)\mathrm{tr}(X^{-1}A)$.
  (iv) Show that det is a submersion.
(b) [Not required] Consider the map

$$f : \mathrm{GL}(n, \mathbb{R}) \to \mathrm{GL}(n, \mathbb{R}), \quad X \mapsto f(X) = X^T X.$$

  (i) Prove: f is smooth, and $df_X(A) = X^T A + A^T X$.
  (ii) Prove: f has constant rank $n(n+1)/2$.

(8) [Not required] [**Germ and the category of pointed manifolds**]
Suppose you are a teacher. Write down lecture notes (no more than 1 page), starting with the definition of germ below (and explore some properties of germ that you may need), then explain what is "the category of pointed smooth manifolds" and why "taking differential at a point" is a functor.

Definition of germ: Let M, N be smooth manifolds and $p \in M$. Let $f_i : U_i \to N$ be smooth maps, where U_i are neighborhoods p. For such

functions, define an equivalence relation by

$$f_1 \sim f_2 \iff \exists U \subset U_1 \cap U_2 s.t. f_1|_U = f_2|_U.$$

The equivalence class containing f is called an *germ* of smooth maps at p and is denoted by $[f]_p$.

Here are some necessary definitions of category and functor you may assume without repeating:

定义 1.148.（范畴）

一个**范畴** $\mathcal{C}$ 包含

1. 一个类 $\mathrm{Ob}(\mathcal{C})$，其中的元素称为 **对象**,
2. 一个类 $\mathrm{Mor}(\mathcal{C})$，其中的元素称为对象间的 **态射**，满足
   - 每个态射 f 都有一个 **始对象** $X \in \mathrm{Ob}(\mathcal{C})$ 和一个 **终对象** $Y \in \mathrm{Ob}(\mathcal{C})$.
     - 我们记 $f: X \to Y$ 并称 “f 是从 X 到 Y 的态射”.
     - 我们将从 X 到 Y 的态射全体记为 $\mathrm{Mor}(X, Y)$. [a]
   - 态射 $f: X \to Y$ 和 $g: Y \to Z$ 的复合是态射 $g \circ f: X \to Z$，且满足

     (a). (结合性) 设 $f: X \to Y, g: Y \to Z$ 和 $h: Z \to W$, 则

     $$h \circ (g \circ f) = (h \circ g) \circ f,$$

     (b). (单位元) 对 $X \in \mathrm{Ob}(\mathcal{C})$, 存在 **单位态射** $\mathrm{Id}_X : X \to X$, 使得对于任意态射 $f: Z \to X$ 和 $g: X \to Y$，都有

     $$\mathrm{Id}_X \circ f = f \quad \text{且} \quad g \circ \mathrm{Id}_X = g.$$

[a] 对于一些对象 X 和 Y, 可能并不存在从 X 到 Y 的态射, 此时 $\mathrm{Mor}(X, Y) = \emptyset$.

定义 3.2.20.（函子）

设 $\mathcal{C}, \mathcal{D}$ 是范畴. 从 $\mathcal{C}$ 到 $\mathcal{D}$ 的 (协变) **函子**[a] F 是一个对应，使得

(1) 将 $\mathrm{Ob}(\mathcal{C})$ 中的每个对象 X 都对应到 $\mathrm{Ob}(\mathcal{D})$ 中的某个对象 $F(X)$,

(2) 将范畴 $\mathcal{C}$ 中的每个态射 $f \in \mathrm{Mor}(X, Y)$ 都对应到范畴 $\mathcal{D}$ 中的某个态射 $F(f) \in \mathrm{Mor}(F(X), F(Y))$, 且该对应满足：

- 对范畴 $\mathcal{C}$ 中的任意对象 $X \in \mathrm{Ob}(\mathcal{C})$，有

$$F(\mathrm{Id}_X) = \mathrm{Id}_{F(X)},$$

- 对范畴 $\mathcal{C}$ 中的任意态射 $f \in \mathrm{Mor}(X, Y)$ 和 $g \in \mathrm{Mor}(Y, Z)$ 都有

$$F(g \circ f) = F(g) \circ F(f).$$

[a] 类似地我们可以定义**反变函子**的概念：反变函子也是一个对应，它将每个态射 $f \in \mathrm{Mor}(X, Y)$ 对应到态射 $F(f) \in \mathrm{Mor}(F(Y), F(X))$ 并且满足 $F(\mathrm{Id}_X) = \mathrm{Id}_{F(X)}$ 和 $F(g \circ f) = F(f) \circ F(g)$.

PROBLEM SET 2, PART 2: REGULAR VALUES
DUE: OCT. 17

(1) [**Measure zero set in smooth manifolds**]
   (a) Prove: the phrase "measure zero" is well-defined on smooth manifolds.
   (b) Deduce Sard's theorem from the Euclidean case.
   (c) Show that if $f : M \to N$ is a smooth map of constant rank $r < \dim N$, then $f(M)$ has measure zero.

(2) [**An counterexample to Sard's Theorem**]
Here is a counterexample to Sard's theorem if f is not smooth enough (constructed by E. Grinberg). Let $C \subset [0,1]$ be the Cantor set.
   (a) Construct a C^1 function $f : \mathbb{R} \to \mathbb{R}$ such that the critical set of f contains C. (Hint: Denote $[0,1] \setminus C = \cup(a_k, b_k)$. Start with a "continuous bump function" f_k on (a_k, b_k) with $\int f_k(t)dt = b_k - a_k$.)
   (b) Show that the function $g : \mathbb{R}^2 \to \mathbb{R}$ defined by $g(x, y) = f(x) + f(y)$ is C^1, and the set of critical values contains an interval. (Hint: Show that $C + C = [0, 2]$.)

(3) [**Morse functions**]
Let $U \subset \mathbb{R}^n$ be an open set, and $f \in C^\infty(U)$.
- We say a critical point $p \in U$ of f is *non-degenerate* if the Hessian matrix

$$\mathrm{Hess} f(p) = \left(\frac{\partial^2 f}{\partial x^i \partial x^j} \right)(p)$$

  is non-degenerate.
- A function is called a *Morse function* if every critical point is non-degenerate.

Prove:
   (a) Use inverse function theorem to prove that non-degenerate critical point must be isolated.
   (b) Given any $f \in C^\infty(U)$, for almost every $a \in \mathbb{R}^n$, the "linear perturbation"

$$f_a : U \to \mathbb{R}, \quad x \mapsto f(x) + a_1 x^1 + \cdots + a_n x^n$$

   of f is a Morse function on U.
   [Hint: Consider regular values of the map $g = df = (\frac{\partial f}{\partial x^1}, \cdots, \frac{\partial f}{\partial x^n}) : U \to \mathbb{R}^n$.]
   (c) (Not required) Suppose U is bounded. Prove: for any $f \in C^\infty(U)$ and any $\varepsilon > 0$, there is a Morse function $g \in C^\infty(U)$ so that $|g - f| < \varepsilon$ and all critical values of g are distinct.
   (d) (Not required) Extend the result in (c) to smooth functions defined on a compact manifold.

(4) [**The Lagrange multiplier**]
Let M be a smooth manifold, and $f \in C^\infty(M)$ a smooth function. We would like to study the critical points of the function $\tilde{f} := f|_S \in C^\infty(S)$ for a smooth submanifold $S \subset M$. For simplicity, we suppose there is a smooth map $g : M \to N$ and a regular value $p \in N$ of g so that $S = g^{-1}(q)$. Prove: a point $p \in S$ is a critical point of $\tilde{f}$ if there exists a linear function $L : T_qN \to \mathbb{R}$, (called a *Lagrange multiplier*), so that $df_p = L \circ dg_p$.

(5) [**Proper maps**]
Recall that a map is called *proper* if the pre-image of any compact set is compact. Let $f : M \to N$ be a smooth and proper map.
(a) Prove: If an injective immersion $f : M \to N$ is proper, then it is an embedding.
(b) Now suppose $\dim M = \dim N$, and suppose $q \in f(M)$ be a regular value of f. Prove: $f^{-1}(q)$ is a finite set $\{p_1, \cdots, p_k\}$, and there exist a neighborhood V of q in N and neighborhoods U_i of p_i in M such that
- $U_1, \cdots, U_k$ are disjoint coordinate charts in M,
- $f^{-1}(V) = U_1 \cup \cdots \cup U_k$,
- For each $1 \leq i \leq k$, f is a diffeomorphism from U_i onto V.

(6) [**The cotangent bundle**]
Let M be a smooth manifold of dimension n. Let T_p^*M be the dual vector space of T_pM, with a dual basis $\{dx^1, \cdots, dx^n\}$ (which is defined locally for a coordinate chart of M) which is defined to be the dual of $\{\partial_1, \cdots, \partial_n\}$. Let $T^*M = \bigcup_p T_p^*M$ be the disjoint union of all T_p^*M. We will call T^*M the *cotangent bundle* of M.
(a) Modify PSet2-1-3 to endow with T^*M a topology so that it is a smooth manifold of dimension $2n$.
(b) Prove: T^*M is orientable.
(c) (Not required) Prove: If f is a smooth function on M, then the map
$$s_f : M \to T^*M, \quad p \mapsto (p, df_p)$$
is an injective immersion and is proper. [In particular, its image is a smooth submanifold of T^*M.]
(d) (Not required) For any $(p, \xi_p) \in T^*M$, the tangent space $T_{(p,\xi_p)}T^*M \simeq T_pM \oplus T_p^*M$.

PROBLEM SET 3, PART 1: EMBEDDING AND NEIGHBORHOODS
DUE: OCT. 31

(1) [**Embedding $\mathbb{T}^2$ into $\mathbb{R}^3$**]
(a) Recall that

$$\mathbb{T}^2 = S^1 \times S^1 = \{(x^1, x^2, x^3, x^4) \mid (x^1)^2 + (x^2)^2 = 1, (x^3)^2 + (x^4)^2 = 1\}.$$

Let $\iota : \mathbb{T}^2 \hookrightarrow \mathbb{R}^4$ be the canonical embedding and $g : \mathbb{R}^4 \to \mathbb{R}^3$ be the map

$$g : \mathbb{R}^4 \to \mathbb{R}^3, \quad (x^1, x^2, x^3, x^4) \mapsto (x^1(2 + x^3), x^2(2 + x^3), x^4).$$

(i) For any $p = (x^1, x^2, x^3, x^4) \in \mathbb{T}^2$, write down a basis of $\mathrm{Im}(d\iota)_p$.
(ii) Write down the matrix of dg.
(iii) Prove: The map $f = g \circ \iota : \mathbb{T}^2 \to \mathbb{R}^3$ is an embedding.

(b) (Not required) Prove: $S^m \times S^n$ can be embedded into $\mathbb{R}^{m+n+1}$.
(Hint: First embed $S^n \times \mathbb{R}$ into $\mathbb{R}^{n+1}$)

(2) [**Embedding of $\mathbb{RP}^2$ into $\mathbb{R}^4$**]
(a) Consider the map

$$f : S^2 \to \mathbb{R}^3, \quad (x, y, z) \mapsto (yz, zx, xy).$$

Prove: It induces an immersion of $\mathbb{RP}^2$ outside several (how many?) points.

(b) Prove: The map

$$f : S^2 \to \mathbb{R}^3, \quad (x, y, z) \mapsto (x^2 - y^2, yz, zx, xy)$$

is an embedding of $\mathbb{RP}^2$ to $\mathbb{R}^4$.

(3) [**The sphere bundle**]
(a) Prove: The "sphere bundle" SM described in the proof of Theorem 2.5.8 is a smooth submanifold of TM of dimension $2m - 1$.
[Hint: Find a smooth function on M so that SM is a regular level set.]

(b) Write down a complete proof of Theorem 2.5.8.

(4) [**The normal bundle**]
(a) Prove Proposition 2.6.6.

(b) (Not required) More generally, Proposition 2.6.6 for the normal bundle $N(X, M)$, where X is a smooth submanifold of M.

(c) (Not required) Suppose $f \in C^\infty(\mathbb{R}^K)$ and 0 is a regular value of f. Let $X = f^{-1}(0)$. Prove: The normal bundle $N(X, \mathbb{R}^K)$ is diffeomorphic to $X \times \mathbb{R}$.

(d) (Not required) Let $\Delta = \{(x, x) \mid x \in M\} \subset M \times M$ be the diagonal. Prove:

$$\Phi : TM \to N(\Delta, M \times M), \quad (x, v) \mapsto ((x, x), (v, -v))$$

is a diffeomorphism.

(5) [**Restriction of smooth map to smooth submanifold**]
Let $f : M \to N$ be a smooth map, $\iota : X \hookrightarrow M$ be a smooth submanifold, Y be a smooth submanifold of N, and $f(X) \subset Y$.
(a) Prove: $\tilde{f} = f \circ \iota : X \to Y$ is a smooth map.
(b) Prove: If f is an immersion on X, then $\tilde{f}$ is an immersion.
(c) Prove: The condition "$f : X \to f(X)$ is a diffeomorphism" in the statement of Theorem 2.6.4 can be weakened to "f is injective on X".

(6) [**Immersions/embeddings are dense**]
(a) Let $U \subset \mathbb{R}^m$ be open, and $f : U \to \mathbb{R}^n$ be smooth. Suppose $n \geq 2m$. Prove: For any $\varepsilon > 0$, there exists an $m \times n$ matrix $A = (a_{ij})$ with $|a_{ij}| < \varepsilon$, such that the map
$$g : U \to \mathbb{R}^n, \quad g(x) = f(x) + Ax$$
is an immersion.
(b) (Not required) Let M be a compact smooth manifold of dimension m, and $f : M \to \mathbb{R}^K$ be a smooth map. Suppose $K \geq 2m$. Prove: For any $\varepsilon > 0$, there is an immersion $f_\varepsilon : M \to \mathbb{R}^K$ with $|f - f_\varepsilon| < \varepsilon$.
(c) (Not required) State and prove a similar theorem for embedding.

(7) [**Isotopy**] (Not required)
Let $f, g : M \to N$ be two embeddings. An *smooth isotopy* between f and g is a smooth map $F : M \times \mathbb{R} \to N$ so that
- $F(x, 0) = f(x), F(x, 1) = g(x)$;
- $F_t := F|_{M \times \{t\}}$ is an embedding for $0 \leq t \leq 1$.

(a) Prove: Smooth isotopy is an equivalence relation.
(b) Prove: If the induced level-preserving map
$$G : M \times \mathbb{R} \to N \times \mathbb{R}, \quad G(x, t) = (F(x, t), t)$$
is an embedding, then $F : M \times \mathbb{R} \to N$ is an isotopy.
(c) Prove: If each F_t is a proper embedding, then the induced level-preserving map G is an embedding. [In particular, the conclusion holds if M is compact.]

PROBLEM SET 3, PART 2: TRANSERSALITY
DUE: OCT. 31

(1) [Smooth submanifolds as regular level sets]
Here is what we argued in class:

> 一方面，由子流形的定义可知，对于任意 $q \in X$，均存在邻域 V 以及由坐标卡映射所诱导的光滑映射 $g : V \to \mathbb{R}^l$（其中 $l = \dim N - \dim X$ 是 X 在 N 中的余维数）使得 $g^{-1}(0) = X \cap V$。注意由 g 的构造可知 dg 是满射，于是 0 是 g 的正则值，且它的核为
>
> $$\ker(dg_q) = T_q X.$$

(a) Summarize the above as a lemma.
(b) Show that the "global version" fails: Let K be the Klein bottle and let S be its central circle. Prove that there is no smooth function $f : K \to \mathbb{R}$ so that 0 is a regular value and $f^{-1}(0) = S$.
(c) (Not required) In general, suppose S is a smooth submanifold of M of codimension r, and suppose the normal bundle $N(S, M)$ is trivial in the sense that $N(S, M)$ is isomorphic(You may need to find the exact meaning of "isomorphic" between vector bundles) to $S \times \mathbb{R}^r$. Prove: There is a smooth map $f : M \to \mathbb{R}^r$ so that $0 \in \mathbb{R}^r$ is a regular value of f and $S = f^{-1}(0)$.
(d) (Not required) Conversely, suppose $f : M \to \mathbb{R}^r$ is a submersion, and $S = f^{-1}(0)$ is a submanifold of M. Show that $N(S, M)$ is trivial.
(e) Write down a theorem summarizing (c) and (d).

(2) [**Stability of various properties**]
We need the following definition:

> We say a property $\mathcal{P}$ concerning maps in $C^\infty(M, N)$ is a *stable property* if it is preserved under small deformation, namely, if $f \in C^\infty(M, N)$ satisfies $\mathcal{P}$ and F is a smooth homotopy with $F(x, 0) = f$, then there exists $\varepsilon > 0$ so that for each $0 < t < \varepsilon$, the map $f_t(\cdot) = F(\cdot, t)$ satisfies the property $\mathcal{P}$.

(a) Prove: If M is compact, then the following properties of maps in $C^\infty(M, N)$ are stable:
  (i) submersion,
  (ii) embedding.
(b) (Not required) Show that the conclusion above fails if M is non-compact by studying the following example.

> **Counterexample:** Let $h \in C^\infty(\mathbb{R})$ be a smooth bump function with
>
> $$h(x) = 1 \text{ for } |x| < 1, \qquad h(x) = 0 \text{ for } |x| > 2.$$
>
> Consider the map $F(x, t) = xh(tx)$ as a homotopy with $f_0(\cdot) = F(\cdot, 0)$.

(3) [**Stability of transversal intersection**]

(a) Show that if M is compact and X is a smooth submanifold of N, then the property "$f \in C^\infty(M,N)$ intersect X transversally" is a stable property.

(b) (Not required) Let $F : S \times M \to N$ be a smooth map. Suppose M is compact, and $X \subset N$ is a closed submanifold. Denote $f_s(\cdot) = F(s,\cdot)$. Prove: the set

$$\{s \in S \mid f_s \text{ intersect } X \text{ transversally}\}$$

is an open subset of S.

(4) [**Lefschetz maps**]

You will need the following conceptions.

Let $f : M \to M$ be a smooth map. A point $p \in M$ is a called a *fixed point* of f if $f(p) = p$. We say f is a *Lefschetz map* if for each fixed point p of f, 1 is not an eigenvalue of $df_p : T_pM \to T_pM$. The *local Lefschetz number* $L_p(f)$ of a Lefschetz map at a fixed point p is the sign of the determinant $\det(df_p - \mathrm{Id})$, i.e. $L_p(f) := 1$ if $\det(df_p - \mathrm{Id}) > 0$, and $L_p(f) := -1$ if $\det(df_p - \mathrm{Id}) < 0$.

Do the following questions:

(1) Let $r_\theta : S^2 \to S^2$ be the map "rotate S^2 by an angle θ", $(\theta \neq 2k\pi)$, defined by

$$r_\theta(x^1, x^2, x^3) = (x^1 \cos\theta - x^2 \sin\theta,\ x^1 \sin\theta + x^2 \cos\theta,\ x^3).$$

Prove: r_θ is a Lefschetz map.

(2) Let V be a vector space, and $L : V \to V$ a linear map. Let $\Delta = \{(v,v) : v \in V\}$ be the diagonal in $V \times V$, and $\Gamma_L = \{(v, Lv) : v \in V\}$ be the graph of L in $V \times V$. Prove: Γ_L intersects Δ transversally if and only if 1 is not an eigenvalue of L.

(3) Prove: If M is compact and $f : M \to M$ is a Lefschetz map, then f has only finitely many fixed points.

(4) The Lefschetz number of a Lefshetz map f is defined to be $L(f) = \sum_{f(p)=p} L_p(f)$, where the summation is over all fixed points p. Compute $L(r_\theta)$for r_θ in (1).

(5) [**Simply connectedness of $\mathbb{R}^n \setminus M$ ($\dim M \leq n-3$)**] (Not required)

Let M be a connected smooth manifold of dimension m. Prove: if $S \subset M$ is a smooth submanifold of dimension $k \leq m-3$, then for any $p \in M - S$, $\pi_1(M,p) \simeq \pi_1(M-S,p)$.

PROBLEM SET 4, PART 1: VECTOR FIELDS
DUE: NOV. 14

(1) [**Coefficients of smooth vector fields**]
Prove Lemma 3.2.6 and Proposition 3.1.7.

(2) [**The commutator of two vector fields**]
Prove Proposition 3.1.13 in three different ways:
(a) **Method 1**: show that in a local chart (φ, U, V), if $X = \sum X^i \partial_i$ and $Y = \sum Y^j \partial_j$, then we have $[X, Y] = \sum_{i,j} (X^i \partial_i Y^j - Y^i \partial_i X^j) \partial_j$, which is a vector field.
(b) **Method 2**: Check that $X_p \circ Y - Y_p \circ X$ is a tangent vector at p which depends smoothly on p.
(c) **Method 3**: Show that $[X, Y]$ is a derivation on $C^\infty(M)$.

(3) [$\Gamma^\infty(TM)$ **as a Lie algebra**]
Prove Proposition 3.1.17.

(4) [**Examples of smooth vector fields**]
Construct smooth vector fields satisfying the given constraints:
(a) A smooth vector field X on S^{2n+1} such that $X_p \neq 0$ for all p.
(b) A smooth vector field X on $\mathbb{T}^n$ such that $X_p \neq 0$ for all p.
(c) A smooth vector field X on S^2 such that $X_p = 0$ for exactly two points.
(d) [Not required] A smooth vector field X on S^2 such that $X_p = 0$ for exactly one point.
(e) Three smooth vector fields on S^3 that are everywhere linearly independent.

(5) [**The sum of two complete vector fields may fail to be complete**]
Let $X_1 = y^2 \frac{\partial}{\partial x}$ and $X_2 = x^2 \frac{\partial}{\partial y}$ be two vector fields on $\mathbb{R}^2$.
(a) Prove: X_1 and X_2 are complete.
(b) Prove: $X_1 + X_2$ is not complete.
[Hint: Consider the integral curve starting at some point (c, c) with $c \neq 0$.]

(6) [**Any derivation on continuous functions**] (Not required)
Can one find a nontrivial map $D : C^0(M) \to C^0(M)$ such that

$$D(fg) = (Df)g + f(Dg)?$$

Justify your conclusion.

(7) [**The Hessian**] (Not required)
(a) Given any $X_p \in T_pM$. Prove: there exists $X \in \Gamma^\infty(TM)$ so that $X(p) = X_p$.

(b) Let $f \in C^\infty(M)$, and suppose p is a critical point of f. Define

$$\mathrm{Hess}_f : T_pM \times T_pM \to \mathbb{R}, \quad (X_p, Y_p) \to Y_p(Xf),$$

where $X \in \Gamma^\infty(TM)$ is any vector field so that $X(p) = X_p$. Prove: Hess_f is well-defined, bilinear and symmetric.

(c) For $M = \mathbb{R}^n$, what is Hess_f?

(d) A critical point p is called non-degenerate if Hess_f is non-degenerate at p. f is called a Morse function if every critical point is non-degenerate. Prove: On any smooth manifold, one can find Morse functions.

(8) [**First integral**] (Not required)

Let X be a smooth vector field on M and let $\gamma_p : J_p \to M$ be the maximal integral curve of X passing p.

(a) Suppose for any $(a, b) \subset J_p$, $\gamma_p((a, b))$ lies in a compact subset of M. Prove: $J_p = \mathbb{R}$.

(b) Suppose $f \in C^\infty(M)$ is a *first integral* of X, i.e. $Xf \equiv 0$. Prove: $\gamma_p \subset f^{-1}(f(p))$.

(c) Suppose X admits a first integral whose level sets are all compact. Prove: X is complete.

PROBLEM SET 4, PART 2: FLOWS, DISTRIBUTIONS
DUE: NOV. 14

(1) [**Plenty of diffeomorphisms**]
Let M be a connected smooth manifold of dimension $m \geq 2$, and let $\{p_1, \cdots, p_n\}$ and $\{q_1, \cdots, q_n\}$ be two sets of points in M.
(a) Prove: there exists a diffeomorphism Φ such that $\Phi(p_i) = q_i$ holds for all i.
(b) Let $X_i \in T_{p_i}M$ and $Y_i \in T_{q_i}N$ $(1 \leq i \leq n)$ be nonzero vectors. Prove: the diffeomorphism Φ above can be chosen so that $d\Phi_{p_i}(X_i) = Y_i$ holds for all i..

(2) [**φ-related vector fields**]
(a) Let $\varphi : M \to N$ be a smooth map. We say two vector fields $X \in \Gamma^\infty(TM)$ and $Y \in \Gamma^\infty(TN)$ are *φ-related*, if for any $p \in M$,

$$d\varphi_p(X_p) = Y_{\varphi(p)}.$$

Prove:
(i) If X and Y are φ-related, then $X(\varphi^* g) = \varphi^*(Yg)$ holds for any $g \in C^\infty(N)$.
(ii) If X_i are φ-related to Y_i for $i = 1, 2$, then $[X_1, X_2]$ is φ-related to $[Y_1, Y_2]$.
(iii) If X and Y are φ-related, $\gamma : I \to M$ is an integral curve of X, then $\varphi \circ \gamma : I \to N$ is an integral curve of Y.
(b) Now let $\varphi : M \to N$ be a diffeomorphism, one can define a "push-forward" operator $\varphi_* : \Gamma^\infty(TM) \to \Gamma^\infty(TN)$ by

$$(\varphi_* X)_{\varphi(p)} = d\varphi_p(X_p).$$

Prove:
(iv) $(\varphi_* X)g = (\varphi^{-1})^* X \varphi^* g$ for any $g \in C^\infty(N)$.
(v) If $X, Y \in \Gamma^\infty(TM)$, then $\varphi_*([X, Y]) = [\varphi_* X, \varphi_* Y]$.

(3) [**Time dependent vector fields**]] (NOT required)
Read Remark 3.3.5. Rewrite it to be a subsection entitled "Time dependent vector fields", including necessary definition(s), theorem(s) and proof(s).

(4) [**The Frobenius condition**]
(a) Let $f, g \in C^\infty(M)$, $X, Y \in \Gamma^\infty(TM)$. Show

$$[fX, gY] = fg[X, Y] + f(Xg)Y - g(Yf)X.$$

(b) Prove Lemma 3.4.11.

(5) [**Commuting vector fields**]
(a) Let $X_1, \cdots, X_k$ be smooth vector fields on M that are pointwise linearly independent on a region W of M, so that $[X_i, X_j] = 0$ for all $1 \leq i, j \leq k$. Prove:

For any point $p \in W$, one can find a local chart (φ, U, V) near p so that $X_i = \partial_i$ on U for all $1 \leq i \leq k$.

(b) Find a coordinate chart on $\mathbb{R}^3 \setminus \{(0,0,z) \mid z \in \mathbb{R}\}$ so that

$$\partial_1 = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}, \quad \partial_2 = x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}, \quad \partial_3 = \frac{\partial}{\partial z}.$$

(c) [Not required] Let $\{\phi_t^X\}$ and $\{\phi_t^Y\}$ be the one-parameter group of diffeomorphisms generated by the vector field X and Y respectively. Prove: $[X, Y] = 0$ if and only if $\phi_t^X \circ \phi_s^Y = \phi_s^Y \circ \phi_t^X$ holds for all $t, s \in \mathbb{R}$.

(6) [**A distribution and its integral manifolds**]
Consider the distribution $\mathcal{V}$ on $\mathbb{R}^3$ spanned by

$$V = x\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + x(y+1)\frac{\partial}{\partial z} \quad \text{and} \quad W = \frac{\partial}{\partial x} + y\frac{\partial}{\partial z}.$$

(a) Show that $\mathcal{V}$ is involutive.

(b) Consider the projection map $\pi : \mathbb{R}^3 \to \mathbb{R}^2, (x, y, z) \mapsto (x, y)$. Show that

$$X = \frac{\partial}{\partial x} + y\frac{\partial}{\partial z} \quad \text{and} \quad Y = \frac{\partial}{\partial y} + x\frac{\partial}{\partial z}$$

are the vector fields spanning $\mathcal{V}$ that are π-related to $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$.

(c) Find the integral curves of X and Y respectively.

(d) What are the integral manifolds of $\mathcal{V}$?

(7) [**Morse theory**] (NOT required)

(a) [**Morse Lemma**] Suppose p is a non-degenerate critical point of a Morse function f (c.f. PSet4-1-7(d)). Prove: There exists a coordinate chart on which f can be written as

$$f(x) = f(p) - x_1^2 - \cdots - x_r^2 + x_{r+1}^2 + \cdots + x_n^2.$$

(b) [**Reeb's theorem**] Suppose f is a Morse function on a closed manifold M of dimension n. If f admits exactly two critical points, then M is homeomorphic to S^n.

(8) [**Foliation**] (NOT required)
Let $\mathcal{F}$ be a collection of disjoint connected immersed k-dimensional submanifolds of M whose union is M. If for any $p \in M$, there is a local chart (φ, U, V) near p so that V is a cube, and each submanifold in $\mathcal{F}$ intersects U in either the empty set, or a countable union of k-dimensional slices of the form $x^{k+1} = c^{k+1}, \cdots, x^m = c^m$, then we say $\mathcal{F}$ is a *foliation* on M, and call the submanifolds in $\mathcal{F}$ the *leaves* of the foliation. Prove:

(a) If $\mathcal{F}$ is a foliation, then the tangent spaces to the leaves of $\mathcal{F}$ form an involutive distribution $\mathcal{V}$ on M.

(b) Conversely, if $\mathcal{V}$ is an integrable distribution, then the maximal connected integral manifolds of $\mathcal{V}$ form a foliation on M.

PROBLEM SET 5, PART 1: LIE GROUPS AND LIE ALGEBRAS
DUE: NOV. 28

(1) [**Left invariant vector fields**]
Let X be a left invariant vector field on a Lie group G. Prove:
(a) X is smooth.
(b) X is complete.

(2) [**The exponential map for matrix Lie groups**]
Define the norm of a matrix $A = (a_{ij})$ to be $\|A\| = (\sum_{i,j} |a_{ij}|^2)^{1/2}$. It is easy to check $\|A + B\| \leq \|A\| + \|B\|$ and $\|AB\| \leq \|A\|\|B\|$.
(a) Show that the exponential map $\exp : M(n, \mathbb{R}) \to \mathrm{GL}(n, \mathbb{R})$ is given by

$$\exp(A) = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots .$$

(you need to prove that the series is absolutely convergent)
(b) [Not required] Let $V = \{A \in M(n, \mathbb{R}) \mid \|I_n - A\| < 1\}$. Define

$$\log : V \to M(n, \mathbb{R}), \quad \log A = -\sum_{k=1}^{\infty} \frac{1}{k}(I_n - X)^k.$$

Prove: log is a smooth map form V to $\mathrm{GL}(n, \mathbb{R})$ and is the local inverse of exp.
(c) [Not required] Prove: For any $A, B \in M(n, \mathbb{R})$,

$$e^{A+B} = \lim_{k\to\infty} (e^{A/k} e^{B/k})^k$$
$$e^{[A,B]} = \lim_{k\to\infty} (e^{A/k} e^{B/k} e^{-A/k} e^{-B/k})^k.$$

(3) [**Lie group structure on S^3**]
Identify S^3 with the set of quaternions of norm 1.
(a) With the help of the multiplication of quaternions, show that S^3 is a Lie group, and determine its Lie algebra.
(b) (Not required) Show that as Lie groups, S^3 is isomorphic to

$$SU(2) = \left\{ \begin{pmatrix} z & w \\ -\bar{w} & \bar{z} \end{pmatrix} \mid z, w \in \mathbb{C}, |z|^2 + |w|^2 = 1 \right\}.$$

(4) [**Abelian Lie groups**]
(a) Let G be a Lie group. Prove: G is abelian (i.e. $g_1 g_2 = g_2 g_1$ for all $g_1, g_2 \in G$) if and only if $\mathfrak{g}$ is abelian (i.e. $[X_1, X_2] = 0$ for all $X_1, X_2 \in \mathfrak{g}$).
(b) (Not required) Prove: Any connected abelian Lie group of dimension n is isomorphic to $\mathbb{R}^r \times \mathbb{T}^{n-r}$ for some r.

(5) [**Simply connected Lie groups**] [Not required]
Show that the universal covering space of a connected Lie group admits a Lie group structure such that the covering map is a Lie group homomorphism.

(6) [**Surjectivity of the exponential map**]
(a) Show that $\exp : \mathfrak{so}(2) \to \mathrm{SO}(2)$ is surjective, and thus $\mathrm{SO}(2)$ is connected.
(b) Show that $\exp : \mathfrak{sl}(2,\mathbb{R}) \to \mathrm{SL}(2,\mathbb{R})$ is not surjective, but $\mathrm{SL}(2,\mathbb{R})$ is connected. [Hint: Show that $\mathrm{tr}(A^2) \geq -2$ for $A \in \mathrm{SL}(2,\mathbb{R})$.]

(7) [**The affine group**]
Consider the smooth manifold $G = \mathrm{GL}(n,\mathbb{R}) \times \mathbb{R}^n$. Define a multiplication operation on G via

$$(X, x) \cdot (Y, y) := (XY, Xy + x).$$

(a) Prove: G is a Lie group with respect to this multiplication.(the *affine group* of $\mathbb{R}^n$)
(b) Consider the vector space $\mathfrak{g} = \mathfrak{gl}(n,\mathbb{R}) \times \mathbb{R}^n$. Define a bracket on $\mathfrak{g}$ via

$$[(A, a), (B, b)] := (AB - BA, Ab - Ba).$$

Prove: $(\mathfrak{g}, [\cdot,\cdot])$ is a Lie algebra.
(c) Define a map $e : \mathfrak{g} \to G$ by

$$e(A, a) := \left(e^A, \sum_{n=1}^{\infty} \frac{1}{n!} A^{n-1} a \right).$$

For any $(A, a) \in \mathfrak{g}$, let $h(t) := e(tA, ta)$. Prove: $\{h(t) \mid t \in \mathbb{R}\}$ is a one-parameter subgroup of G.
(d) [Not required] Find a subgroup of $\mathrm{GL}(n+1,\mathbb{R})$ that is isomorphic to G, and explain how do we get $\mathfrak{g}$ and e above.

(8) [**Lie algebras of more linear Lie groups**]
For any $n \times n$ matrix B, we define

$$\mathrm{GL}_B(n,\mathbb{R}) = \{X \in \mathrm{GL}(n,\mathbb{R}) \mid X^T B X = B\}.$$

(a) Prove: $\mathrm{GL}_B(n,\mathbb{R})$ is a Lie subgroup of $\mathrm{GL}(n,\mathbb{R})$.
(b) Find the Lie algebra $\mathfrak{gl}_B(n,\mathbb{R})$ of $\mathrm{GL}_B(n,\mathbb{R})$.

(9) [**No small subgroup**] [Not required]
Show that any Lie group G does not have small subgroup, i.e. prove that there exists an open neighborhood U of $e \in G$ such that $\{e\}$ is the only subgroup of G that is entirely contained in U.

PROBLEM SET 5, PART 2: LIE GROUPS ACTIONS
DUE: NOV. 28

(1) [**The Lie algebra $\mathfrak{so}(3)$**]
(a) Prove: The Lie algebra $\mathfrak{so}(3)$ has a basis

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Moreover, prove it is isomorphic to the Lie algebra $(\mathbb{R}^3, \times)$, where $\times$ is the cross product.
(b) Consider the $\mathrm{SO}(3)$-action on $\mathbb{R}^3$ by rotations. Prove: The induced vector fields associated to A, B, C on $\mathbb{R}^3$ are

$$x^3\frac{\partial}{\partial x^2} - x^2\frac{\partial}{\partial x^3}, \qquad x^1\frac{\partial}{\partial x^3} - x^3\frac{\partial}{\partial x^1}, \qquad x^2\frac{\partial}{\partial x^1} - x^1\frac{\partial}{\partial x^2}.$$

(2) [**Invariants of a Lie group action**] [Not required]
Suppose G acts on M smoothly.
(a) Prove: The formula

$$(g \cdot \varphi)(x) := \varphi(g^{-1} \cdot x)$$

defines a linear action of G on $C^\infty(M)$.

Definition 0.1. A fixed point of the above induced action of G on $C^\infty(M)$ is called an *invariant* of the G-action on M. The set of invariants is denoted by $C^\infty(M)^G$.

(b) Prove: If G is connected, then $f \in C^\infty(M)^G$ if and only if $X_M f = 0$ for all $X \in \mathfrak{g}$.
(c) Suppose G acts on M properly and freely. Prove: $C^\infty(M)^G \simeq C^\infty(M/G)$.
(d) What are the invariants for the standard $\mathrm{O}(n)$ action on $\mathbb{R}^n$?
(e) What are the invariants for the adjoint action of $\mathrm{GL}(n, \mathbb{C})$ on $\mathfrak{gl}(n, \mathbb{C})$?

(3) [**More Lie subgroups/subalgebras**]
Let G is a Lie group, with Lie algebra $\mathfrak{g}$.
(a) Define the center of G to be $Z(G) = \{g \in G \mid gg' = g'g \text{ for all } g' \in G\}$. Show that $Z(G)$ is a Lie subgroup of G, and find its Lie algebra.
(b) Let $\varphi : G \to G$ be a Lie group homomorphism. Let H be the set of φ-fixed points, i.e. $H = \{g \in G \mid \varphi(g) = g\}$. Prove that H is a Lie subgroup of G, and find its Lie algebra.

(4) [**Equivariant maps**] [Not required]
Let M, N be G-manifolds. A smooth map $f : M \to N$ is called G-equivariant if

$$g \cdot f(m) = f(g \cdot m), \forall g \in G, m \in M.$$

Prove:

(a) Prove: If G acts transitively on M, then f is of constant rank.
(b) Suppose G is connected, then f is G-equivariant if and only if it is $\mathfrak{g}$-equivariant in the sense

$$X_N(f(m)) = df_m(X_m), \forall X \in \mathfrak{g}, m \in M.$$

PROBLEM SET 6, PART 1: TENSORS AND DIFFERENTIAL FORMS
DUE: DEC. 14

(1) [**Tedious multi-linear algebra**]
Prove Lemma 5.1.19 and Proposition 5.1.21.

(2) [**Decomposable tensors**]
A tensor $T \in V_1 \otimes V_2 \otimes \cdots \otimes V_k$ is called *decomposable* if there exists $v_1 \in V_1, \cdots, v_k \in V_k$ so that $T = v_1 \otimes \cdots \otimes v_k$. [Indecomposable tensors are known as *entangled states* in quantum information theory.]
   (a) Let v_1, v_2 be linearly independent vectors in V. Show that the 2-tensor $v_1 \otimes v_1 + v_2 \otimes v_2 \in V \otimes V$ is not decomposable.
   (b) Let V, W be vector spaces, with basis $\{v_i\}$ and $\{w_j\}$ respectively. Prove: A nonzero $T = \sum_{ij} a_{ij} v_i \otimes w_j \in V \otimes W$ is decomposable if and only if the coefficient matrix (a_{ij}) has rank 1.
   (c) [not required] For any vector space V, we let $\mathbb{P}(V) = V - \{0\}/\{v \sim \lambda v\}$ be the projective space of V. Prove: The map

$$\mathbb{P}(V_1) \times \cdots \times \mathbb{P}(V_k) \to \mathbb{P}(V_1 \otimes \cdots \otimes V_k), \quad ([v_1], \cdots, [v_k]) \mapsto [v_1 \otimes \cdots \otimes v_k]$$

   is an embedding. This is known as the (generalized) *Segre embedding*.
   (d) Suppose $k \geq 2$, and $\dim V_i \geq 2$ for each $1 \leq i \leq k$. Prove: Any decomposable k-tensor $T \in V_1 \otimes \cdots \otimes V_k$ lines in the image of the generalized Segre embedding above. Conclude that most k-tensors are indecomposable.

(3) [**Tensor product of vector spaces**]
Let V, W be finite dimensional vector spaces, with basis $\{v_i\}$ and $\{w_j\}$ respectively.
   (a) Prove: $V \otimes W^* \simeq L(W, V)$ as vector spaces. Then answer the following problems:
      (i) What is the trace of $T \in V \otimes V^* \simeq L(V, V)$?
      (ii) Give a conceptional proof of the fact that the contraction defined in Definition 5.1.5 is well-defined.
   (b) [not required] **[Universal Property]** Prove: There exists a blinear map $\varphi : V \times W \to V \otimes W$, so that the following *universal property* holds: for any vector space Z and any bilinear map $h : V \times W \to Z$, there exists a unique linear map $\tilde{h} : V \otimes W \to Z$ so that $h = \tilde{h} \circ \varphi$. Moreover, such a pair $(V \otimes W, \varphi)$ is unique up to isomorphism.

(4) [Lie derivative on differential forms] [Not required]
   (a) Read section 5.2.3.
   (b) Prove Proposition 5.2.11.

(5) [**Maxwell's equations**]

The famous *Maxwell's equations* in the theory of electromagnetism are of the form

$$\nabla \cdot \vec{B} = 0, \qquad \frac{\partial \vec{B}}{\partial t} + \nabla \times \vec{E} = 0,$$
$$\nabla \cdot \vec{E} = 4\pi\rho, \qquad \frac{\partial \vec{E}}{\partial t} - \nabla \times \vec{B} = -4\pi\vec{J},$$

where $\vec{E} = \langle E_x, E_y, X_z\rangle$ is the electricity field, $\vec{B} = \langle B_x, B_y, B_z\rangle$ is the magnetism field, ρ is the charge density, and $\vec{J} = \langle J_x, J_y, J_z\rangle$ is the current density. All functions, e.g. E_x, E_y, B_x, ρ, J_x, are functions of x, y, z and t.

Define the *Faraday 2-form* F on $\mathbb{R}^4$ (with coordinates t, x, y, z) as

$$\mathbf{F} = E_x dx\wedge dt + E_y dy\wedge dt + E_z dz\wedge dt + B_x dy\wedge dz + B_y dz\wedge dx + B_z dx\wedge dy,$$

and define the current 1-form to be

$$\mathbf{J} = \rho dt + J_x dx + J_y dy + J_z dz.$$

We also define the *Hodge star operators* on $\mathbb{R}^4$ (with respect to the *Minkowski metric*) to be the linear operators

$$* : \Omega^k(\mathbb{R}^4) \to \Omega^{4-k}(\mathbb{R}^4), \qquad 0 \le k \le 4,$$

so that on 2-forms one has $** = -1$ and

$$*(dx\wedge dt) = dy\wedge dz, \qquad *(dy\wedge dt) = -dx\wedge dz, \qquad *(dz\wedge dt) = dx\wedge dy,$$

while on 1-forms one has $dt\wedge(*dt) = dx\wedge(*dx) = \cdots = dt\wedge dx\wedge dy\wedge dz$, e.g.

$$*dt = dx\wedge dy\wedge dz, \quad *dx = -dt\wedge dy\wedge dz, \quad *dy = dt\wedge dx\wedge dz \quad \textit{etc.}$$

Prove: The system of Maxwell equations is equivalent to

$$d\mathbf{F} = 0, \qquad d*\mathbf{F} = 4\pi * \mathbf{J}.$$

(6) [**The canonical symplectic form on** $\mathbb{R}^{2n}$]

Denote by $(x^1,\cdots,x^n,y^1,\cdots,y^n)$ the coordinate functions on $\mathbb{R}^{2n}$. Let $\omega = \sum_{i=1}^{n} dx^i\wedge dy^i$.

(a) What is $d\omega$?
(b) What is $\omega^n = \omega\wedge\cdots\wedge\omega$ (wedge n times)?
(c) Find $\iota^*\omega$ for the following three cases:
  (i) $\iota : \mathbb{R}^{2n-2} \hookrightarrow \mathbb{R}^{2n}$ be the embedded submanifold of $\mathbb{R}^{2n}$ defined by $x^n = y^n = 0$.
  (ii) $\iota : \mathbb{R}^n \hookrightarrow \mathbb{R}^{2n}$ be the submanifold of $\mathbb{R}^{2n}$ defined by $y^1 = \cdots = y^n = 0$.
  (iii) $\iota : \mathbb{T}^n \hookrightarrow \mathbb{R}^{2n}$ be the submanifold of $\mathbb{R}^{2n}$ defined by $(x^i)^2 + (y^i)^2 = 1, 1 \le i \le n$.
(d) For any $f \in C^\infty(\mathbb{R}^{2n})$, find a vector field X_f so that $\iota_{X_f}\omega = df$.
(e) Let X_f be as above. Prove: $\mathcal{L}_{X_f} f = 0$ and $\mathcal{L}_{X_f}\omega = 0$.

(7) [**Involutive distributions revisited**]

(a) Let $\omega = ydx - dz$ be a smooth 1-form on $M = \mathbb{R}^3$. For each $p \in M$, let $\mathcal{V}_p := \ker(\omega|_{T_pM})$. Prove: $\mathcal{V}$ is a distribution of rank 2 on M. Is it integrable?

(b) Let $\mathcal{V}$ be a smooth distribution of rank k on M. Prove: For each $p \in M$, there exists a neighborhood U of p and smooth 1-forms $\omega_1, \cdots, \omega_{n-k}$ on U such that for each $q \in U$, $\mathcal{V}_q = \ker(\omega_1|_{T_qM}) \cap \cdots \cap \ker(\omega_{n-k}|_{T_qM})$.

Definition 0.1. We say a differential form $\eta \in \Omega^j(M)$ annihilates a smooth distribution $\mathcal{V}$ if $\eta(X_1, \cdots, X_j) = 0$ for all $X_i \in \mathcal{V}$.

(c) [Not required] Let $\mathcal{V}$ be a smooth distribution of rank k on M, and $\omega_1, \cdots, \omega_{n-k}$ are as described above. Prove: $\eta \in \Omega^j(M)$ annihilates $\mathcal{V}$ if and only if there exists $(j-1)$-forms $\alpha_1, \cdots, \alpha_{n-k}$ so that $\eta = \sum_{i=1}^{n-k} \omega_i \wedge \alpha_i$.

(d) [Not required] Prove: a smooth distribution $\mathcal{V}$ is involutive if and only if the following criteria is satisfied:

> If η is a 1-form that annihilates $\mathcal{V}$ on an open set U, then $d\eta$ also annihilates $\mathcal{V}$ on U.

(e) [Not required] Let $\omega_1, \cdots, \omega_{n-k}$ be as in (b). Prove: $\mathcal{V}$ is involutive if and only if there exists 1-forms α^i_j $(1 \le i, j \le n-k)$ so that

$$d\omega^i = \sum_{j=1}^{n-k} \omega^j \wedge \alpha^i_j, \qquad 1 \le i \le n-k.$$

(Ref: John Lee, *Introduction to Smooth Manifolds* page 493-495.)

PROBLEM SET 6, PART 2: INTEGRALS ON MANIFOLDS
DUE: DEC. 14

(1) [**Orientable manifolds**]
   (a) Prove: Any Lie group G is orientable.
   (b) [not required] Suppose M is orientable, $f : M \to N$ is smooth, and $q \in N$ is a regular value of f. Prove: $f^{-1}(q)$ is orientable.

(2) [**Orientation-preserving diffeomorphism**]
   (a) Characterize orientation-preserving diffeomorphism via volume forms.
   (b) Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be the map $f(x) = -x$. Is f orientation-preserving?
   (c) Let $f : S^n \to S^n$ be the antipodal map $f(x) = -x$. Is f orientation-preserving?
   (d) Prove: For any k, $\mathbb{RP}^{2k}$ is not orientable.

(3) [**An integral**]
Consider the 2-form

$$\omega = xdy \wedge dz + y^2 dz \wedge dx + zdx \wedge dy$$

on $\mathbb{R}^3$. Let $\iota : S^2 \to \mathbb{R}^3$ be the standard embedding. Compute the integral $\int_{S^2} \iota^* \omega$.

(4) [**The divergence of a smooth vector field**]
Let M be a smooth manifold, μ a volume form on M, and X a smooth vector field on M. Then the Lie derivative $\mathcal{L}_X \mu$ is again a smooth n-form on M. The *divergence* of X with respect to the volume form μ is the function $\mathrm{div}_\mu(X)$ such that

$$\mathcal{L}_X \mu = \mathrm{div}_\mu(X)\mu.$$

   (a) Find the expression of $\mathrm{div}_\mu(X)$ in local coordinates.
   (b) Prove: For any $f \in C^\infty(M)$,

$$\mathrm{div}_\mu(gX) = g\mathrm{div}_\mu(X) + X(g)\mu.$$

   (c) Prove **Gauss formula**:

$$\int_M (\mathrm{div}_\mu X)\mu = \int_{\partial M} \iota_X \mu.$$

   (d) We say a vector field X is *divergence free* if $\mathrm{div}_\mu(X) = 0$. Prove: If M is a compact manifold without boundary (i.e. $\partial M = \emptyset$), and X is a divergence free vector field on M, then for any $f, g \in C^\infty(M)$, one has

$$\int_M X(f)g\mu = -\int_M fX(g)\mu.$$

   (In other words, the linear operator $X : C^\infty(M) \to C^\infty(M)$ is skew-symmetric.)

(5) [**The transport equation**] [not required]
Let M be a smooth manifold, μ a volume form on M, and $\Omega \subset M$ an open set in M with smooth boundary. Let X be a complete vector field on M, and $\Omega_t = \phi_t(\Omega)$ the flow-out of Ω by the flow ϕ_t generated by X.

(a) For any $\rho \in C^\infty(\mathbb{R} \times M)$ (as usual we denote $\rho_t = \rho(t, \cdot) \in C^\infty(M)$), prove:

$$\frac{d}{dt} \int_{\Omega_t} \rho_t \mu = \int_{\Omega_t} \left(\frac{\partial \rho}{\partial t} + \mathrm{div}_\mu(\rho_t X) \right) \mu. \qquad \text{[The transport equation]}$$

(b) Let ρ be the mass density of an ideal fluid moving in a compact region. Assume the *conservation of mass*, i.e. for any region Ω, $\int_{\Omega_t} \rho_t \mu$ is independent of t. Prove:

$$\frac{\partial \rho}{\partial t} + \mathrm{div}_\mu(\rho_t X) = 0. \qquad \text{[The equation of continuity]}$$

PROBLEM SET 7, PART 1: DE RHAM COHOMOLOGY GROUPS
DUE: DEC. 28

(1) [**An infinite dimensional H^1_{dR}**]
   (a) Find a closed 1-form on $\mathbb{R}^2 - \{(0,0)\}$ that is not exact.
   (b) Let $M = \mathbb{R}^2 - \mathbb{Z}^2$. Prove: $H^1_{dR}(M)$ is infinitely dimensional.

(2) [**The de Rham cohomologies of the union**]
   (a) Let $M = M_1 \cup M_2$ be the disjoint union of two smooth manifolds. Find the relation between $H^k_{dR}(M)$ and $H^k_{dR}(M_i)$.
   (b) Calculate the de Rham cohomology groups of S^1 via Mayer-Vietoris sequence.
   (c) What if $M = \cup_{i=1}^{\infty} M_i$ is the disjoint union of countably many smooth manifolds?

(3) [**Witten deformation**]
Let M be a smooth manifold, and $\theta \in \Omega^1(M)$ be an exact 1-form.
   (a) For any k, define $d_\theta : \Omega^k(M) \to \Omega^{k+1}(M)$ by

$$d_\theta(\omega) = d\omega + \theta \wedge \omega.$$

   Prove: $d_\theta(d_\theta \omega) = 0$ for all $\omega \in \Omega^k(M)$.
   (b) Consider the complex

$$0 \to \Omega^0(M) \xrightarrow{d_\theta} \Omega^1(M) \xrightarrow{d_\theta} \cdots \xrightarrow{d_\theta} \Omega^{m-1}(M) \xrightarrow{d_\theta} \Omega^m(M) \to 0.$$

   Define $Z^k_\theta(M)$, $B^k_\theta(M)$ and $H^k_\theta(M)$, and prove: $H^k_\theta(M)$ is isomorphic to $H^k_{dR}(M)$.

(4) [**Missing parts in the proof of Mayer-Vietoris sequence**]
   (a) Prove Proposition 6.2.1.
   (b) (NOT required) To prove Theorem 6.2.4, we need to prove six inclusion relations. We proved one in class. Try to prove the rest five relations.

(5) [**Applications of Mayer-Vietoris sequence**]
   (a) Let M_1, M_2 be connected manifolds, and $M = M_1 \# M_2$ be their connected sum. Find the relation between $H^k_{dR}(M)$ and $H^k_{dR}(M_1), H^k_{dR}(M_2)$.
   (b) Compute the de Rham cohomology groups of $\mathbb{RP}^2$ and $\mathbb{CP}^2$.
   (c) (NOT required) Compute the de Rham cohomology groups of $\mathbb{RP}^n$ and $\mathbb{CP}^n$.

(6) [**Kunneth formula and its applications**]
   (a) Read §6.2.2 "Application 3: Kunneth formula".
   (b) Prove $H^k_{dR}(\mathbb{T}^n) \simeq \mathbb{R}^{\binom{n}{k}}$.

(c) For any compact manifold M, define $p_M(t)$ to be the polynomial $p_M(t) = \sum_{i=0}^{n} b_i(M)t^i$. Prove: If M, N are compact, then

$$p_{M\times N}(t) = p_M(t)p_N(t).$$

(In particular, we get $\chi(M \times N) = \chi(M)\chi(N)$.)

(d) (NOT required) Prove: $S^m \times S^n$ is homeomorphic to $S^{m'} \times S^{n'}$ if and only if $\{m, n\} = \{m', n'\}$.

(7) **[Not required]** Let G be a compact connected Lie group acting smoothly on a smooth manifold M (from left).

(a) For each k, define left-invariant k-forms on M, and then define the "k^{th} left-invariant de Rham cohomology group" $H^k_L(M)$.

(b) Prove $H^k_L(M) \simeq H^k_{dR}(M)$, as follow:

(i) Let $i : \Omega^k_L(M) \hookrightarrow \Omega^k_{dR}(M)$ be the inclusion map . Prove: i induces a linear map $i_* : H^k_L(M) \to H^k_{dR}(M)$.

(ii) Prove i_* is injective.

Hint: Let dg be a normalized Haar measure on G, in other words, dg is the measure associated to a volume form α on G which is both left and right invariant, such that $\int_G \alpha = 1$. [You may try to prove the existence of such dg if you want.] For each $\omega \in \Omega^k(M)$ define the averaging of ω with respect to G to be

$$A(\omega) = \int_G \tau_g^*\omega \; dg.$$

Show that A is a linear map from $\Omega^k(M)$ to $\Omega^k_L(M)$, which induces a linear map $A_* : H^k_{dR}(M) \to H^k_L(M)$. Moreover, prove $A_* \circ i_* = \mathrm{Id}$.

(iii) Prove: i_* is surjective.

Hint: It's enough to prove $[A(\omega)] = [\omega]$ for any $[\omega] \in H^k_{dR}(M)$. First notice that the map A above can be rewritten as

$$A(\omega) = \int_G \tau^*\omega \wedge \pi^*\alpha,$$

where $\tau : G \times M \to M$ is the action of G on M, and $\pi : G \times M \to G$ is the projection, and one regards the differential form $\tau^*\omega \wedge \pi^*\alpha$ as a top form on G (with M variables as parameters). Take a contractible neighborhood U of e in G, and a top form β on G which is supported in U and satisfies $\int_G \beta = 1$. Then there exists a differential form η on G so that $\alpha - \beta = d\gamma$. (Why?) Let $\tau_U : U \times M \to M$ be the restriction of τ. Then $\tau^*\omega \wedge \pi^*\beta = \tau_U^*\omega \wedge \pi^*\beta$. Finally prove $\tau_U^*\omega = \pi_M^*\omega + d\eta$ for some differential form η on $U \times M$, where $\pi_M : U \times M \to M$ is the projection.

(c) Prove: $H^k_{dR}(\mathbb{T}^n) \simeq \mathrm{span}\{dx^{i_1} \wedge \cdots \wedge dx^{i_k} \mid 1 \le i_1 < \cdots < i_k \le n.\}$.

PROBLEM SET 7, PART 2: DE RHAM COHOMOLOGY GROUPS
DUE: DEC. 28

(1) [**The de Rham cohomology groups of punctured manifolds**]
   (a) Compute $H^1_{dR}(\mathbb{T}^2 - \{p\})$.
   (b) Suppose $n > 1$. Compute $H^k_{dR}(\mathbb{R}^n - \{p, q\})$ and $H^k_c(\mathbb{R}^n - \{p, q\})$.
   (c) Let M be a compact oriented connected smooth manifold of dimension $m > 1$, $p \in M$.
      (i) Prove: $H^m_{dR}(M \setminus \{p\}) = 0$.
      (ii) For $k < m$, prove: $H^k_{dR}(M \setminus \{p\}) = H^k_{dR}(M)$.

(2) [**Invariance of compactly supported de Rham cohomology groups**]
Prove Theorem 1.3.3 and corollary 1.3.5.

(3) [**The mapping degree**]
   (a) Let $f : S^1 \to S^1$ be the map $f(\theta) = e^{ik\theta}$. Find $\deg(f)$.
   (b) Let $f : \mathbb{T}^n \to \mathbb{T}^n$ be given by $f(e^{i\theta_1}, \cdots, e^{i\theta_n}) = (e^{ik_1\theta_1}, \cdots, e^{ik_n\theta_n})$. Find $\deg(f)$.
   (c) Prove: If $f : S^m \to S^m$ has degree $\deg(f) \neq (-1)^m$, then it admits at least one fixed point.
   (d) Prove: Any smooth map $f : S^2 \to \mathbb{T}^2$ has degree 0.

(4) [**The fundamental theorem of algebra**]
   (a) Let $f : \mathbb{C} \to \mathbb{C}$ be a polynomial of degree $k > 0$. Prove: f is proper, and is properly homotopic to $f_1(z) = z^k$.
   (b) Use mapping degree to prove the fundamental theorem of algebra.

(5) [**Orientation covering**] (NOT required)
Let M be a connected non-orientable manifold of dimension m. Let

$$\widetilde{M} = \{(p, O_p) \mid p \in M, O_p \text{ is an orientation on } T_pM\}.$$

Let $\pi : \widetilde{M} \to M$ be the obvious projection map.
   (a) Show that $\widetilde{M}$ admits a natural topological and smooth structure so that it is an m-dimensional connected orientable manifold, and π is a smooth double covering map.
   (b) Show that the non-trivial Deck transformation $\sigma : \widetilde{M} \to \widetilde{M}$ is an orientation-reverting diffeomorphism.
   (c) Find the relation between $H^k_{dR}(M)$ and $H^k_{dR}(\widetilde{M})$.

(6) [**The Euler characteristics**]

(a) Let B be a compact smooth manifold with boundary ∂B. Let $2B$ be the manifold obtained by "gluing" two copies of B along the boundary. Prove:

$$\chi(2B) + \chi(\partial B) = 2\chi(B).$$

(b) (NOT required) Given the following formula (the answer to a problem in previous PSet)

$$H^k_{dR}(\mathbb{RP}^{2n}) = \begin{cases} \mathbb{R}, & k = 0, \\ 0, & k \neq 0. \end{cases}$$

Prove: $\mathbb{RP}^{2n}$ can not be the boundary of any $2n+1$ dimensional compact manifold.

(7) [H^1_{dR} **v.s.** π_1] (NOT required)

Let M be a compact connected smooth manifold. Consider the map

$$\Psi : H^1_{dR}(M) \to \mathrm{Hom}(\pi_1(M), \mathbb{R}), \qquad [\omega] \mapsto \left([\gamma] \mapsto \int_\gamma \iota_\gamma^* \alpha \right)$$

where we only take smooth representatives $\gamma \in [\gamma]$, and the map $\iota_\gamma : \gamma \hookrightarrow M$ is the inclusion map.

(a) Prove: The map Ψ is well-defined.
[Hint: use the cochain homotopy property.]

(b) Prove: The map Ψ is injective.
[In fact the map Ψ is also surjective, but the proof is hard.]
In particular, if M is simply connected, then $H^1_{dR}(M) = 0$.

(8) [**Moser's trick**] (NOT required)

Let M be a m-dimensional compact orientable manifold, and $\omega_0, \omega_1 \in \Omega^m_{dR}(M)$ are two volume forms on M such that $\int_M \omega_0 = \int_M \omega_1$. Show that there exists a diffeomorphism $\varphi : M \to M$ such that $\varphi^* \omega_2 = \omega_1$ as follows:

(a) For $t \in [0,1]$, let $\omega_t = (1-t)\omega_1 + t\omega_2$. Prove: ω_t is a family of volume forms on M, and there exists $\alpha \in \Omega^{m-1}(M)$ so that $\dot{\omega}_t = d\eta$ holds for all $t \in [0,1]$.

(b) Let X_t be the time-depending vector field so that $\iota_{X_t}\omega_t = -\alpha$. Let φ_t be the flow of X_t. Prove: $\varphi_t^* \omega_t = \omega_0$ holds for all $t \in [0,1]$.